在定量社会科学研究中负责任地使用人工智能

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研究方向：运用因果推断与计算方法评估数字信息技术的社会与政治影响

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计算研究中的转变

……（计算技能）门槛降低

……专业知识的重新分配

……“对学生而言，问题不再是他们能否编程，而是他们能否审视、批判、调整并负责任地部署 AI 辅助方法，而不把便利误当作……”

the ease of generation raises the value of human judgment in CCR further, necessitating sharper and more transparent human oversight.

This redistribution of expertise has major implications for pedagogy and training. For instructors, GenAI unsettles the sequencing of computational education itself: it is no longer obvious whether CCR methods courses should begin with traditional techniques and only later introduce generative systems, or whether GenAI should be integrated from the outset as part of the field's ordinary methodological environment. For students, the question is no longer whether they can code, but whether they can audit, critique, adapt, and responsibly deploy AI-assisted methods without mistaking convenience

for competence.

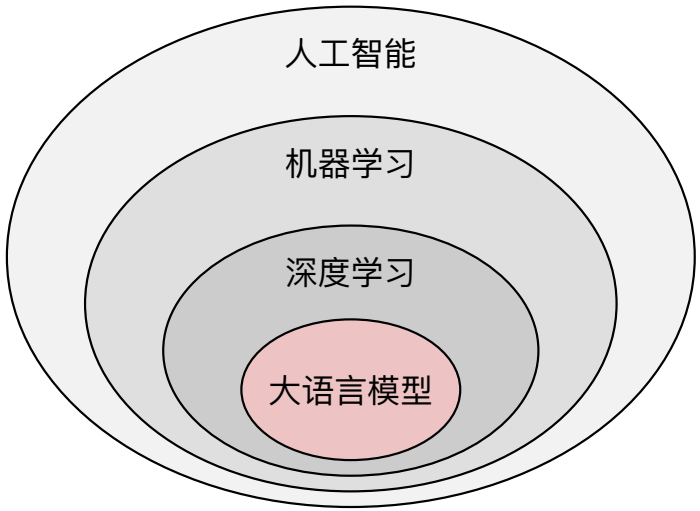
At the same time, the seeming democratization of computational workflows may conceal new inequalities. Well-resourced scholars with strong methodological priors are often best positioned to benefit from GenAI because they can afford to access the most advanced models and the computational power to run them. Furthermore, they may be better equipped to

来源：Zhou et al. (2026)（第 4–5 页）

人工智能：宏观图景

什么是人工智能?

- “制造智能机器的科学与工程”
  (McCarthy)
- 能够**感知**环境并为实现目标而**行动**的系统
- 一个层层嵌套的技术家族，而非单一技术



人工智能简史

- 观察：人工智能是一个已发展数十年的综合性领域
  - **1950 年代**图灵测试（1950）；达特茅斯会议首次提出“人工智能”（1956）
  - **1960–80 年代**符号主义人工智能与专家系统；两次“AI 寒冬”
  - **1990–2000 年代**统计机器学习：从数据中学习规则
  - **2012** 深度学习在图像识别上取得突破
  - **2017** Transformer 架构问世
  - **2018–2020** 预训练语言模型（BERT、GPT-2、GPT-3）
  - **2022–** ChatGPT；**通用大语言模型**进入日常科研实践

前大模型时代的定量研究 AI

前大模型时代 AI 在定量社会科学研究中的应用

- **监督学习**：人工标注样本，训练分类器，再推广到全样本
- **无监督学习**：主题模型、聚类、词嵌入
- **文本、图像与网络即数据** (King et al., 2013)
- **机器学习用于因果推断**：异质性效应、高维控制变量

共同特点：针对特定任务的模型、依赖标注数据、由研究者搭建流程

and computational linguistics, and later in the social sciences and digital humanities—developed tools, models, and software that facilitated the analysis and organization of texts at scale.

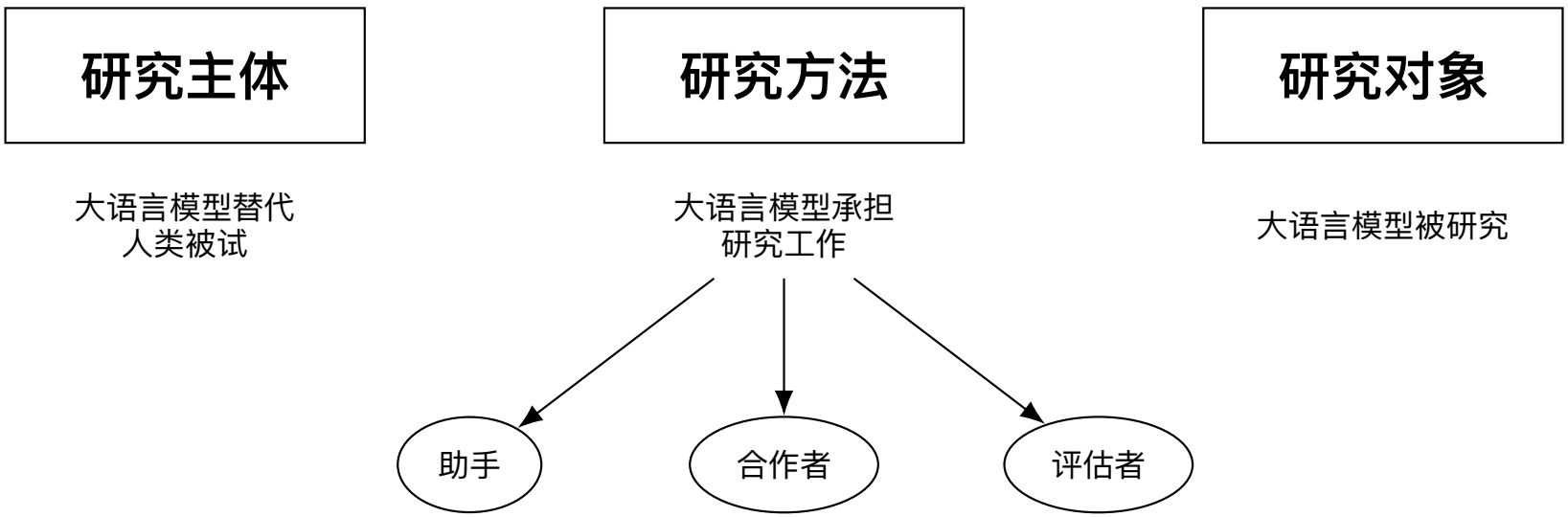
Almost all of the applications of large-scale text analysis in the social sciences use algorithms either first developed in computer science or built closely on those developments. For example, numerous papers within political science—including many of our own—build on topic models (Blei, Ng, and Jordan, 2003; Quinn et al., 2010; Grimmer, 2010; Roberts et al., 2013) or use supervised learning algorithms for document classification (Joachims, 1998; Jones, Wilkerson, and Baumgartner, 2009; Stewart and Zhukov, 2009; Pan and Chen, 2018; Barberá et al., 2021). Social scientists have also made methodological contributions themselves, and in this book we will showcase many of these new models designed to accomplish new types of tasks. Many of these contributions have even flowed from the social sciences to computer science. Statistical models used to analyze roll call votes, such as Item Response Theory models, are now used in several computer science articles (Clinton, Jackman, and Rivers, 2004; Gerrish and Blei, 2011; Nguyen et al., 2015). Social scientists have broadly adapted the tools and techniques of computer scientists to social science questions.

However, the knowledge transfer from computer science and related fields has created confusion in how text as data models are applied, how they are validated, and how their output is interpreted. This confusion emerges because tasks in academic computer science are different than the tasks in social science, the digital humanities, and even parts of industry. While computer scientists are often (but not exclusively!) interested in information retrieval,

来源：Grimmer et al. (2022)（第 23 页）

大语言模型与定量研究

大语言模型在计算社会科学中的角色

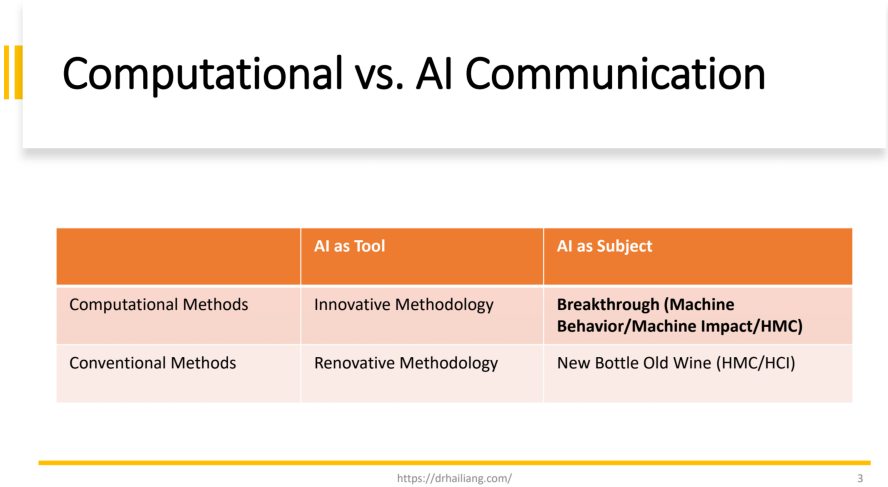


来源：四种角色（研究主体、助手、合作者、评估者）的分类见 Xu et al. (2026)；两种角色（研究工具、研究对象）改编自 Zhou et al. (2026)

AI 作为研究主体

- **硅基样本**：以人物设定为条件的模拟问卷受访者
- **模拟实验**：正式施测前预演设计、检验统计功效
- **基于智能体的仿真**：大量大语言模型智能体相互作用

关键问题：模拟的回答何时能够反映真实的人类行为?



来源：https://drhailiang.com/

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大语言模型作为研究方法：助手

- 编写、调试和转换分析代码
- 检索与总结文献
- 编辑与润色文字
- 数据清洗与整理

研究者设定任务，大语言模型加速执行

大语言模型作为研究方法：合作者

- 头脑风暴研究问题与假设
- 评议研究设计与识别策略
- 提出测量方式、稳健性检验和替代解释
- 智能体工作流：在研究者监督下完成多步骤分析

大语言模型贡献想法，研究者承担责任

大语言模型作为研究方法：评估者

- **标注**：大规模零样本与少样本文本分类
- **大语言模型评审**：对质量、立场或相似度打分
- **审阅**：对稿件、代码和可复现性提供反馈

必须以人工编码的基准进行验证

大语言模型作为研究对象

- **模型审计**：政治倾向、社会偏见与价值对齐
- **对人的影响**：说服、学习、劳动与福祉
- **对信息生态的影响**：AI 生成内容与搜索

局限性

局限性

1. **效度。**输出未必测量了我们想测量的内容
2. **可复现性。**闭源模型、静默更新与随机输出
3. **偏差。**训练数据对部分群体和语言代表性过高
4. **幻觉。**捏造事实与文献
5. **伦理。**隐私、知情同意以及向第三方传输数据
6. **过度依赖。**技能退化与未经审视的默认设置

展望

展望：负责任地使用

- **披露**：模型、版本、提示词、参数和使用日期
- **验证**：大规模应用前与人工基准比对
- **复现**：固定模型版本；尽可能使用开放权重模型
- **担责**：研究设计、推断和结论由人负责

从工具到研究伙伴——但须有护栏

致谢

感谢…。文责自负。

参考文献

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